Exchange Draft

PDF of the report is included on the next page

Questions:

- Is there a positive relationship between pesticide use and pesticide contamination of surface waters?

- Which pesticide category has the most positive correlation between estimated use and surface water concentration?

- Was anything in this report unclear, difficult to find, or difficult to interpret?

- Please provide two specific suggestions for improvement

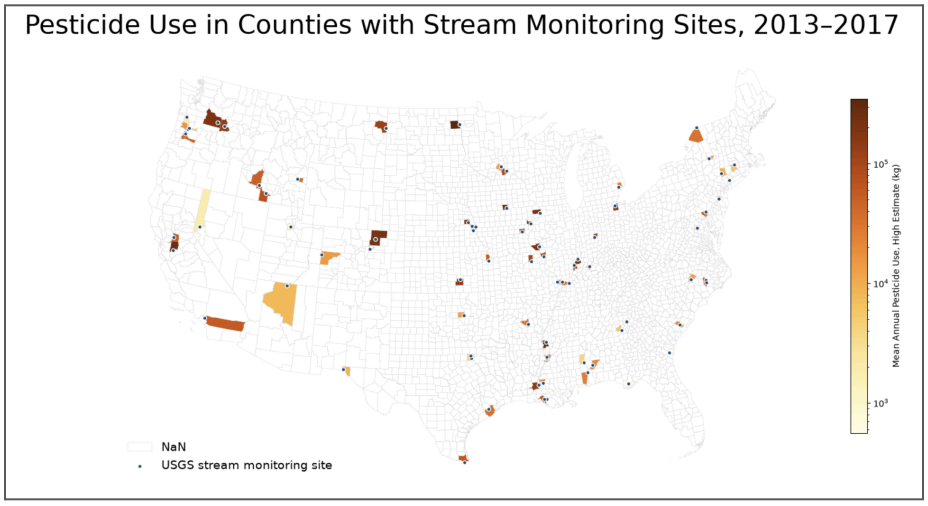
712 Analytical Report Draft

Introduction

This research focuses on the central question of whether counties with higher pesticide use also experience higher concentrations of pesticide contamination in surface water. To further explore this relationship, a supporting question examines whether the strength of this correlation differs among pesticide categories, including herbicides, fungicides, and insecticides. The research is intended for residents concerned about pesticide use and water quality in their communities, farmers, and pesticide companies interested in understanding the environmental impacts of pesticide use.

Results

To create a viable dataset, we combined USGS estimated county pesticide use with USGS surface water sampling locations. To understand the extent of pesticide use within the sampled counties, we generated a choropleth map showing the high estimated value of pesticide use in counties containing a sampling location (Figure 1).



	Value
Water samples	176787
Monitoring sites	79
Counties	74
Pesticides	49
Years	2013–2017

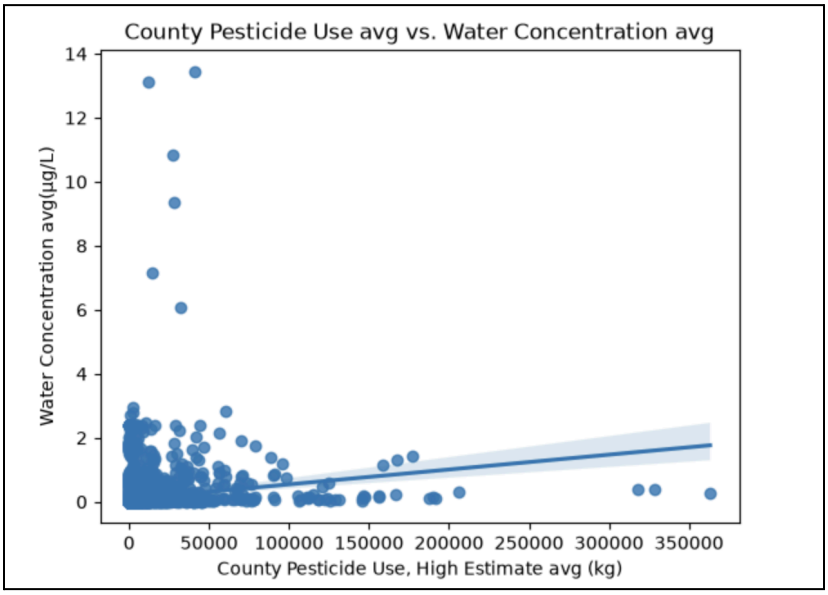
Figure 1

This figure provided a visual representation of the counties within our data, along with the extent of their estimated pesticide use. The associated summary table revealed that among the

74 counties sampled, 49 different pesticides were detected within the 176,787 water samples collected.

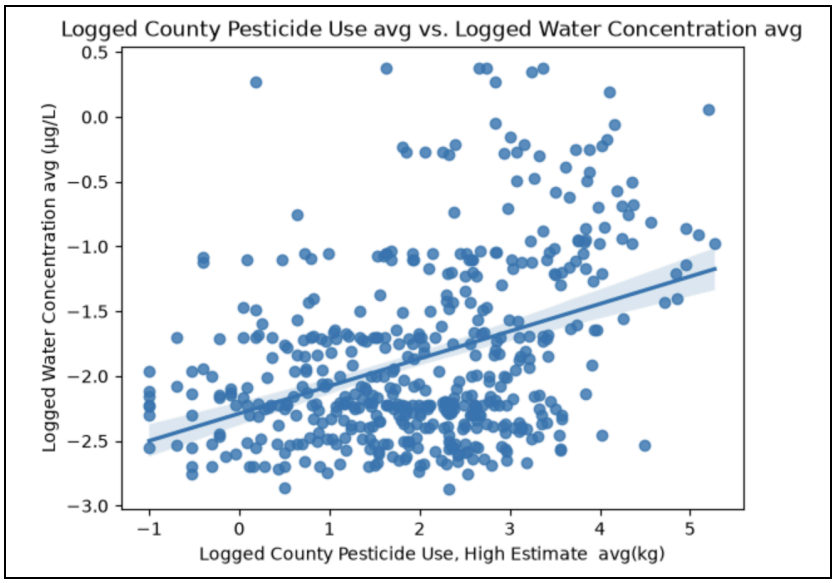
For our main analytical approach, we created a linear regression model. We felt that this approach was best suited for our data, as we are investigating the relationship between two variables. The visual representation of a linear regression clearly asserts whether or not a trend exists within the data, and can be supported with additional statistics. In this case, we are using the R^2 and p-value to determine the strength of the hypothesized relationship.

To address our central question, we began by plotting the estimated pesticide use (kg) by pesticide concentrations (µg/L) in surface waters. As evidenced by Figure 2A and its summary statistics, this initial plot did not demonstrate a notable relationship between pesticide use and contamination. To more effectively investigate the potential relationship between these two variables, we log scaled both axes (Figure 2B).



slope	0.000005
intercept	0.082335
R^2	0.020283
p-value	0.000000

Figure 2A



slope	0.21109
intercept	-2.28161
R^2	0.13747
p-value	0.00000

Figure 2B

A p-value of <0.001 suggests that there is some correlation between pesticide usage and contamination. An R^2 value of 0.14 indicates that county pesticide usage explains 14% of the variation in pesticide concentrations in surface waters. While this suggests that a majority of the variation in surface waters is driven by other factors, such as rainfall, watershed, topography, soil characteristics, etc., it does account for a portion of the contamination variation. A limitation of this figure is that it does not give any information about the type of pesticides being analyzed.

To address our supporting question, we plotted a linear regression for each primary pesticide category: herbicide, fungicide, insecticide, and other (Figure 3).



Figure 3

This visualization revealed that herbicides have a significantly stronger correlation with water contamination than the other pesticides. In tandem with Figure 2, we can now make the assumption that the correlation between county pesticide use and surface water contamination is heavily influenced by herbicide use. To investigate if these relationships underwent significant change throughout the 5 years of sample collection, we iterated this plot through each year in the dataset, 2013-2017. With little change in each pesticide category, this analysis revealed that these trends have been consistent over time.

Finally, to set the stage for further investigation of this topic, we looked into the frequency of occurrence of each pesticide category within our data over time. This plot simply displays how often each pesticide category was included in a water sample and is not representative of the total mass of estimated use.

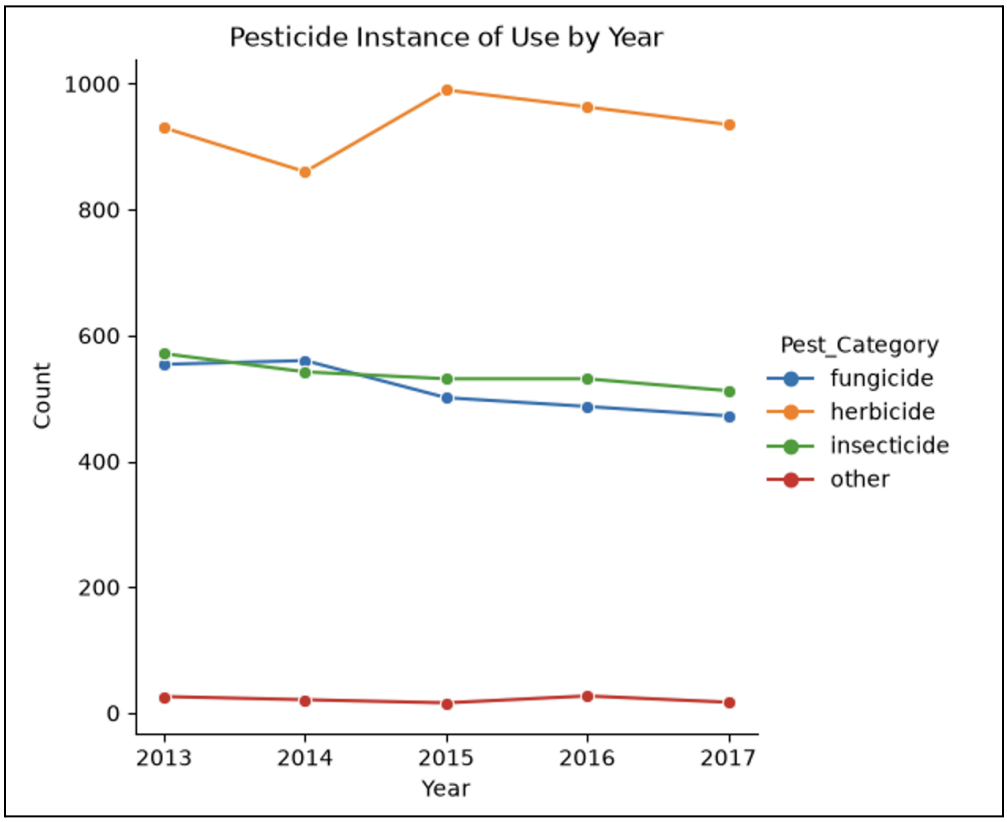


Figure 4

As expected, this plot demonstrates that herbicides were the most commonly sampled pesticide category throughout the 5 years of sample collection. While this does not describe the amount of herbicides used, it does indicate that frequency of use could be a factor in the correlation between herbicide use and contamination. An important limitation of this investigation is the previously calculated R^2 value, which indicates that pesticide usage explains only 14% of the variation in contaminant concentrations. Looking forward, future studies could dive deeper into this relationship to determine how the correlation of these variables is impacted

by frequency/volume of use versus other factors, such as higher persistence in the environment and watershed topography.